

TAURANGA, New Zealand

WMO: 931860

Lat: 37.6706S

Lon: 176.1964E

Elev: 5

StdP: 101.27

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.3	4.2	0.0	3.8	6.2	1.1	4.1	6.6	11.8	13.0	10.7	13.3	2.5	220	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.2	25.8	18.9	24.8	18.7	23.8	18.4	21.1	23.3	20.4	22.7	19.8	22.2	5.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.4	15.1	22.2	19.6	14.4	21.6	18.9	13.7	21.2	61.3	23.3	58.9	22.9	56.8	22.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.3	8.3	1.4	28.4	0.8	1.3	0.8	29.3	0.4	30.1	-0.1	30.8	-0.7	31.7		
(5)			0.5	22.2	0.8	0.8	0.0	22.8	-0.5	23.3	-1.0	23.8	-1.6	24.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	19.7	19.9	18.5	16.0	13.7	11.3	10.6	11.0	12.5	14.1	15.9	18.2	(6)
(7)		DBStd	3.85	1.96	1.85	1.84	2.20	2.33	2.33	2.14	1.93	1.99	1.86	2.05	1.86	(7)
(8)		HDD10.0	49	0	0	0	0	2	13	19	12	3	1	0	0	(8)
(9)		HDD18.3	1342	9	6	21	75	143	211	241	228	174	131	77	26	(9)
(10)		CDD10.0	1904	300	278	263	179	117	51	37	42	79	128	177	253	(10)
(11)		CDD18.3	156	51	51	25	4	0	0	0	0	0	0	3	21	(11)
(12)		CDH23.3	320	130	117	29	3	0	0	0	0	0	0	11	30	(12)
(13)		CDH26.7	15	7	6	1	0	0	0	0	0	0	0	1	1	(13)
(14)	Wind	WSAvg	3.8	4.1	3.8	3.6	3.3	3.4	3.5	3.6	4.0	4.5	4.5	4.2	(14)	
(15)	Precipitation	PrecAvg	1667	108	111	139	161	147	170	175	161	132	117	106	139	(15)
(16)		PrecMax	2203	375	327	296	356	381	284	383	337	237	256	321	323	(16)
(17)		PrecMin	1218	31	16	26	12	19	66	38	84	50	37	36	29	(17)
(18)		PrecStd	234	68	73	68	87	74	53	89	56	46	49	58	74	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	27.5	25.5	23.2	20.4	17.8	16.9	17.3	19.6	20.9	24.7	25.7	(19)
(20)		0.4%	MCWB	19.5	19.8	18.5	17.9	17.0	15.4	14.0	13.3	14.8	15.1	17.3	18.2	(20)
(21)		2%	DB	26.0	25.7	24.1	21.7	19.3	16.9	15.9	16.2	18.0	19.6	22.4	24.1	(21)
(22)		2%	MCWB	18.8	19.4	18.1	17.5	16.5	14.9	13.2	12.8	14.1	14.7	16.4	17.8	(22)
(23)		5%	DB	24.7	24.7	23.1	20.7	18.6	16.2	15.2	15.4	17.1	18.7	21.1	22.9	(23)
(24)		5%	MCWB	18.9	19.0	18.0	17.1	16.0	14.1	12.7	12.5	13.6	14.3	16.0	17.5	(24)
(25)		10%	DB	23.6	23.8	22.3	20.0	17.8	15.4	14.5	14.7	16.2	17.8	19.9	21.8	(25)
(26)		10%	MCWB	18.5	18.9	17.7	16.7	15.3	13.3	12.1	12.1	13.2	14.0	15.4	17.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	22.6	20.8	20.1	18.6	16.8	15.5	15.1	16.1	16.7	18.6	20.3	(27)
(28)		0.4%	MCDWB	24.0	24.4	22.5	21.3	19.3	17.3	16.0	15.9	17.6	18.5	21.4	22.4	(28)
(29)		2%	WB	20.9	21.6	20.2	19.1	17.7	15.7	14.3	14.1	15.2	16.0	17.8	19.6	(29)
(30)		2%	MCDWB	23.4	23.5	22.0	20.4	18.7	16.4	15.3	15.2	16.8	17.9	20.3	21.7	(30)
(31)		5%	WB	20.2	20.8	19.5	18.3	16.9	14.7	13.3	13.2	14.6	15.4	17.2	19.0	(31)
(32)		5%	MCDWB	22.9	23.0	21.6	19.9	18.2	15.8	14.7	14.7	16.3	17.5	19.7	21.2	(32)
(33)		10%	WB	19.5	20.0	18.8	17.4	15.7	13.6	12.4	12.5	13.9	14.9	16.5	18.4	(33)
(34)		10%	MCDWB	22.4	22.5	21.2	19.4	17.3	15.1	14.1	14.2	15.8	17.0	19.1	20.6	(34)
(35)	Mean Daily Temperature Range		MDBR	7.4	7.2	7.5	7.5	7.1	7.0	7.1	7.2	7.2	7.4	6.9	(35)	
(36)		5% DB	MCDDBR	9.2	8.8	8.4	7.6	6.9	6.0	6.7	7.4	7.8	8.1	8.9	8.5	(36)
(37)			MCWBR	4.5	4.2	4.3	4.4	4.5	4.4	4.8	4.9	4.9	4.9	4.7	4.4	(37)
(38)			5% WB	MCDDBR	6.3	6.1	5.9	5.3	5.1	5.0	5.6	5.8	6.2	6.4	6.8	5.8
(39)		MCWBR		3.7	3.8	3.8	4.0	4.1	4.6	4.8	4.6	4.6	4.6	4.3	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.349	0.348	0.328	0.313	0.309	0.302	0.297	0.302	0.329	0.336	0.335	0.352	(40)	
(41)		taud	2.544	2.565	2.620	2.655	2.624	2.622	2.639	2.631	2.525	2.518	2.539	2.530	(41)	
(42)		Ebn at Noon	984	961	943	899	840	817	846	897	925	963	991	985	(42)	
(43)		Edh at Noon	108	101	89	75	67	63	65	75	96	105	108	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.59	5.67	4.71	3.35	2.36	1.85	2.07	2.84	3.85	5.08	6.05	6.24	(44)	
(45)		RadStd	0.43	0.49	0.29	0.24	0.14	0.14	0.15	0.16	0.31	0.34	0.36	0.51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.38	N/A	N/A	N/A	N/A	N/A	-13	-96	+130	+42
(47) Regional (4 neighbors)		+0.30	+0.37	+0.46	N/A	N/A	N/A	-10	-79	+93	N/A

Nomenclature: See separate page